

Perturbed relaxed density, Lagrangian, and Z-vector for the $2n+1$ MP2 second derivatives

PyCC derivation notes

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Companion to `derivints.pdf`. Spin-orbital basis, antisymmetrized two-electron integrals $\langle pq||rs \rangle$. Index convention: i, j, k occupied; a, b, c virtual; p, q, r, s, t all MOs. $\varepsilon_p \equiv f_{pp}$.

These notes are kept *method-general*: D and Γ are the method's *unrelaxed* one- and two-particle correlation densities, and D^{rel} the relaxed one-particle density (unrelaxed plus the orbital-relaxation / Z-vector blocks). The generalized-Fock Lagrangian, the orbital Hessian, and the Z-vector are the standard Gauss–Stanton–Bartlett orbital-response formalism, identical in structure for MP2 and CC (CCSD, CC3, ...); only the densities D, Γ (and, at CC, the amplitude Lagrange multipliers Λ) differ. Points where MP2 *specializes* the general result are flagged as **MP2**: throughout. The MP2 case is what we validate numerically; the CC extension is then a density swap, not a re-derivation.

The starting point is the relaxed-density gradient (`derivints.pdf` Eq. 1, GSB form),

$$\partial_x E = \sum_{pq} \gamma_{pq} f_{pq}^{(x)} + \sum_{pqrs} \Gamma_{pqrs} \langle pq||rs \rangle^{(x)} + \sum_{pq} I_{pq} S_{pq}^{(x)}, \quad \gamma \equiv D^{\text{rel}}, \quad (1)$$

with skeleton (fixed-MO-coefficient) derivatives $f^{(x)}, \langle \rangle^{(x)}, S^{(x)}$. Its second derivative ($2n+1$) needs only *first-order* responses; the new machinery is the response of the relaxed density, which requires the perturbed Z-vector.

Available first-order derivatives. $\partial_x \langle pq||rs \rangle$ and $\partial_x f_{pq}$ are the full (CPHF-folded) MO derivatives (CPHF.`perturbed_eri`, `perturbed_fock`); $\partial_x \varepsilon_p = \partial_x f_{pp}$. $\partial_x D_{pq}$ and $\partial_x \Gamma_{pqrs}$ are the unrelaxed density responses, obtained from the perturbed amplitudes (and, at CC, perturbed Λ). U^x is the first-order orbital-rotation matrix (CPHF.`full_U`); z is the unperturbed Z-vector. **MP2**: the unrelaxed density responses are `MPwfn.perturbed_densities`, built from the closed-form perturbed doubles t^x ; the amplitude Lagrange multipliers coincide with the amplitudes ($\Lambda = t$), so $\partial_x \Lambda = t^x$ needs no separate solve.

1 Unperturbed pieces (for reference)

The generalized-Fock Lagrangian, orbital Hessian, and Z-vector below are method-general (they depend on the method only through D and Γ). **MP2**: the unrelaxed correlation densities are $D_{ij} = -\frac{1}{2} t_{imef} t_{jmef}$, $D_{ab} = +\frac{1}{2} t_{mnbe} t_{mnae}$ (no ov block), and $\Gamma_{ijab} = \frac{1}{4} t_{ijab}$ (oovv/vvoo).

The generalized-Fock Lagrangian (D, Γ inputs):

$$I'_{pq} = -\frac{1}{2} \left[\varepsilon_p (D_{pq} + D_{qp}) + \delta_{q \in \text{occ}} \sum_{rs} D_{rs} (\langle rp||sq \rangle + \langle rq||sp \rangle) + 4 \sum_{rst} \langle pr||st \rangle \Gamma_{qrst} \right]. \quad (2)$$

The occupied–virtual antisymmetric part is the Z-vector right-hand side, $X_{ia} = I'_{ia} - I'_{ai}$, and the orbital Hessian is

$$A_{ia,jb} = \langle aj||ib \rangle + \langle ab||ij \rangle + (\varepsilon_a - \varepsilon_i) \delta_{ij} \delta_{ab}. \quad (3)$$

Solving $\sum_{jb} A_{ia,jb} z_{jb} = X_{ia}$ gives the Z-vector, and $D_{ai}^{\text{rel}} = D_{ia}^{\text{rel}} = -z_{ai}$ (with $D_{ij}^{\text{rel}} = D_{ij}$, $D_{ab}^{\text{rel}} = D_{ab}$). The energy-weighted density is $W = I'(D^{\text{rel}})$.

2 Perturbed Lagrangian $\partial_x I'_{pq}$

Differentiating Eq. (2) term by term (product rule), using the *unrelaxed* D, Γ and their responses:

$$\begin{aligned} \partial_x I'_{pq} = & -\frac{1}{2} \left[(\partial_x \varepsilon_p)(D_{pq} + D_{qp}) + \varepsilon_p(\partial_x D_{pq} + \partial_x D_{qp}) \right. \\ & + \delta_{q \in \text{occ}} \sum_{rs} \left(\partial_x D_{rs} (\langle rp||sq \rangle + \langle rq||sp \rangle) + D_{rs} (\partial_x \langle rp||sq \rangle + \partial_x \langle rq||sp \rangle) \right) \\ & \left. + 4 \sum_{rst} \left(\partial_x \langle pr||st \rangle \Gamma_{qrst} + \langle pr||st \rangle \partial_x \Gamma_{qrst} \right) \right]. \end{aligned} \quad (4)$$

3 Perturbed Z-vector z^x

Differentiating $Az = X$ gives $Az^x = X^x - A^x z$, i.e. the *same* orbital Hessian Eq. (3) with a perturbed right-hand side:

$$\sum_{jb} A_{ia,jb} z_{jb}^x = X_{ia}^x - (A^x z)_{ia}, \quad (5)$$

$$X_{ia}^x = \partial_x I'_{ia} - \partial_x I'_{ai}, \quad (6)$$

$$(A^x z)_{ia} = \sum_{jb} (\partial_x \langle aj||ib \rangle + \partial_x \langle ab||ij \rangle) z_{jb} + \sum_b \partial_x f_{ab} z_{ib} - \sum_j \partial_x f_{ij} z_{ja}. \quad (7)$$

The last two terms are ∂_x of the $(\varepsilon_a - \varepsilon_i)\delta$ block written in its general non-canonical form $f_{ab}\delta_{ij} - f_{ij}\delta_{ab}$; they use the full $\partial_x f$ virtual–virtual and occupied–occupied blocks, not merely $\partial_x \varepsilon$. Then $z^x = A^{-1}(X^x - A^x z)$.

4 Perturbed relaxed density $\partial_x D_{pq}^{\text{rel}}$

The relaxed density is the unrelaxed density plus the orbital relaxation, in every block:

$$D_{pq}^{\text{rel}} = D_{pq} + D_{pq}^{\text{orb}}, \quad D_{ai}^{\text{orb}} = D_{ia}^{\text{orb}} = -z_{ai}, \quad D_{ij}^{\text{orb}} = D_{ab}^{\text{orb}} = 0, \quad (8)$$

so, differentiating,

$$\partial_x D_{ij}^{\text{rel}} = \partial_x D_{ij}, \quad \partial_x D_{ab}^{\text{rel}} = \partial_x D_{ab}, \quad \boxed{\partial_x D_{ai}^{\text{rel}} = \partial_x D_{ai} - z_{ai}^x} \quad (\partial_x D_{ia}^{\text{rel}} = \partial_x D_{ia} - z_{ai}^x). \quad (9)$$

The occupied–virtual block carries *both* the unrelaxed ov response and the perturbed Z-vector. **MP2:** the unrelaxed one-particle density has no occupied–virtual block (no singles: $f_{ai} = 0$ by Brillouin, and the first-order wavefunction is doubles-only), so $D_{ai} = 0$ and $\partial_x D_{ai} = 0$, leaving $\partial_x D_{ai}^{\text{rel}} = -z_{ai}^x$ — the ov derivative is the perturbed Z-vector alone. For CC (CCSD, CC3, ...) singles are present, so $D_{ai} \neq 0$ and the $\partial_x D_{ai}$ term in Eq. (9) survives; the CC extension supplies it (and $\partial_x D_{ij}$, $\partial_x D_{ab}$) from $\partial_x t$ and $\partial_x \Lambda$, with Eqs. (4)–(7) unchanged.

5 Assembly: field second derivative (polarizability)

For an electric field the skeleton Fock derivative is $f^{(b)} = -\mu_b$ and $\langle \rangle^{(b)} = S^{(b)} = 0$, so differentiating the gradient in direction a leaves two terms ($f^{(ab)} = 0$ for a field):

$$\alpha_{ab} = \sum_{pq} \partial_a D_{pq}^{\text{rel}} (\mu_b)_{pq} + \sum_{pq} D_{pq}^{\text{rel}} [(U^a)^\top \mu_b + \mu_b U^a]_{pq}. \quad (10)$$

The second term is the MO dipole rotating under the field, $\partial_a f^{(b)} = \text{rotate}(U^a, -\mu_b)$. Numerically Eq. (10) reproduces the explicit-route polarizability to $\sim 10^{-15}$ (H₂O/6-31G, spin-orbital).

6 Frozen core

The frozen-core gradient recipe (Baeck, Watts, and Bartlett, *J. Chem. Phys.* **107**, 3853 (1997), Eqs. 22–28) supplies the orbital-response density in the dropped-MO case; here we differentiate it. The correlation energy is no longer invariant to core $\leftrightarrow$ active-occupied rotations, so the occupied–occupied block of D^{rel} acquires a core–active part. Writing c for a frozen-core and i for an active-occupied orbital (so D_{ci}^{rel} is a sub-block of the general D_{ij}^{rel} of Sec. 4, refining it across the frozen/active partition), the canonical perturbed-orbital condition (`derivints.pdf` Eq. 14) gives a direct energy-gap divide in place of the unrelaxed value:

$$D_{ci}^{\text{rel}} = \frac{I'_{ci} - I'_{ic}}{\varepsilon_c - \varepsilon_i} = D_{ic}^{\text{rel}}, \quad (11)$$

where the unrelaxed $D_{ci} = 0$ (the frozen core is uncorrelated), so this block — like the ov block — is pure orbital relaxation. It also couples into the ov Z-vector right-hand side X .

Differentiating Eq. (11) requires care. The gap divide is the *canonical* form of the underlying Sylvester relation

$$\sum_d f_{cd} D_{di}^{\text{rel}} - \sum_j D_{cj}^{\text{rel}} f_{ji} = I'_{ci} - I'_{ic}, \quad (12)$$

which reduces to Eq. (11) only because the unperturbed Fock is diagonal ($f_{cd} = \varepsilon_c \delta_{cd}$, $f_{ji} = \varepsilon_i \delta_{ji}$). Differentiating the Sylvester relation and evaluating at the (diagonal) unperturbed Fock gives

$$(\varepsilon_c - \varepsilon_i) \partial_x D_{ci}^{\text{rel}} = \partial_x I'_{ci} - \partial_x I'_{ic} - \sum_d \partial_x f_{cd} D_{di}^{\text{rel}} + \sum_j D_{cj}^{\text{rel}} \partial_x f_{ji}. \quad (13)$$

The diagonal terms ($d = c$, $j = i$) reproduce the naive quotient-rule shift $-D_{ci}^{\text{rel}}(\partial_x \varepsilon_c - \partial_x \varepsilon_i)$; the *off-diagonal* $j \neq i$ active–active (and $d \neq c$ core–core) pieces of $\partial_x f$ are essential and are *not* captured by differentiating the gap alone — a field leaves the active occupied space non-canonical ($\partial_x f_{ji} \neq 0$ off-diagonal), so these couple different orbitals within the block. Dropping them leaves a residual error (H₂O/6-31G frozen core: the polarizability is wrong by $\sim 7 \times 10^{-7}$; with Eq. (13) it recovers the finite-difference value to $\sim 10^{-12}$). The perturbed ov Z-vector right-hand side X^x additionally picks up the derivative of the core–active coupling term. These are folded in for the frozen-core path.